

Hawkins-Bradru-Kass data

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The Hawkins-Bradru-Kass (HBK) data is an artificial data set (Hawkins et al., 1984) that challenges outlier detection methods based on perturbing a single observation. Figure 1 illustrates the pairwise relationships between the three predictor variables X_1, X_2, X_3 and outcome Y in the HBK data. The observations can be clustered into 3 groups: group A, which constitutes the majority of the observations, shows no relationship between the predictor variables and the outcome; group B consists of observations that have high-leverage with respect to group A but maintain the same null relationship with Y ; group C are all outliers that pull the fitted least squares line away from the null.

1 Outlier detection

The masking effect of multiple outliers makes the detection of the outlying group impossible using univariate methods that consider only one observation at a time. Thomas et al. (2018) show how the 3 clusters of observations can be detected after fitting a Bayesian linear model by applying multivariate analysis methods to the posterior variance matrix V of the log likelihood contributions.

In a frequentist analysis, outlier-robust linear regression methods can be used to identify group C as a set of outliers and, after down-weighting their influence, reveal the underlying null relationship in groups A and B. In a Bayesian analysis, one way to improve robustness of a linear model is to use a 2-component normal mixture for the outcome variable Y . This allows a small proportion of the data to be contaminated with observations with a distinct mean and variance from the rest. Details of the mixture model are given in the supplementary materials. After fitting the mixture model, figure 2 shows the loadings for the first principal component of V (top panel) and Ω (bottom panel). The top panel shows the direction of the perturbation ϵ that is maximally influential. This perturbation puts the highest loads on observations in group C. The bottom panel

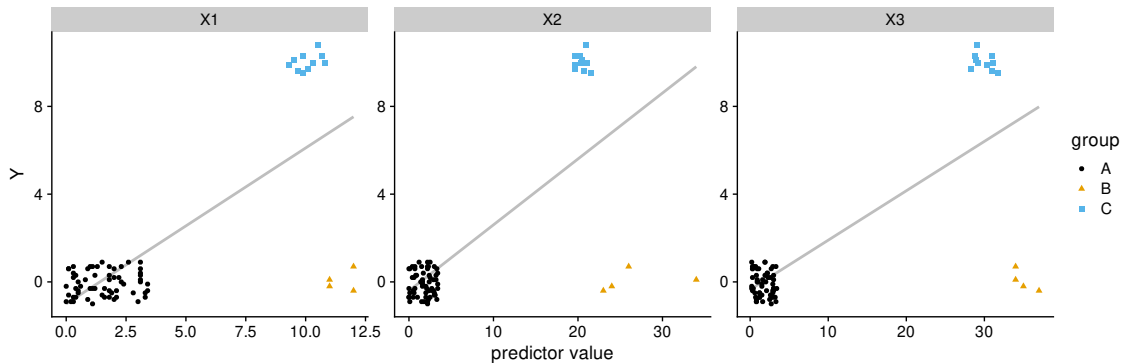


Figure 1: Hawkins-Bradru-Kass (HBK) artificial data set. Group A contains the majority of the data; group B is a set of 4 observations with high leverage; and group C is a set of 10 outliers. The gray line shows a univariate least-squares fit to each predictor variable.

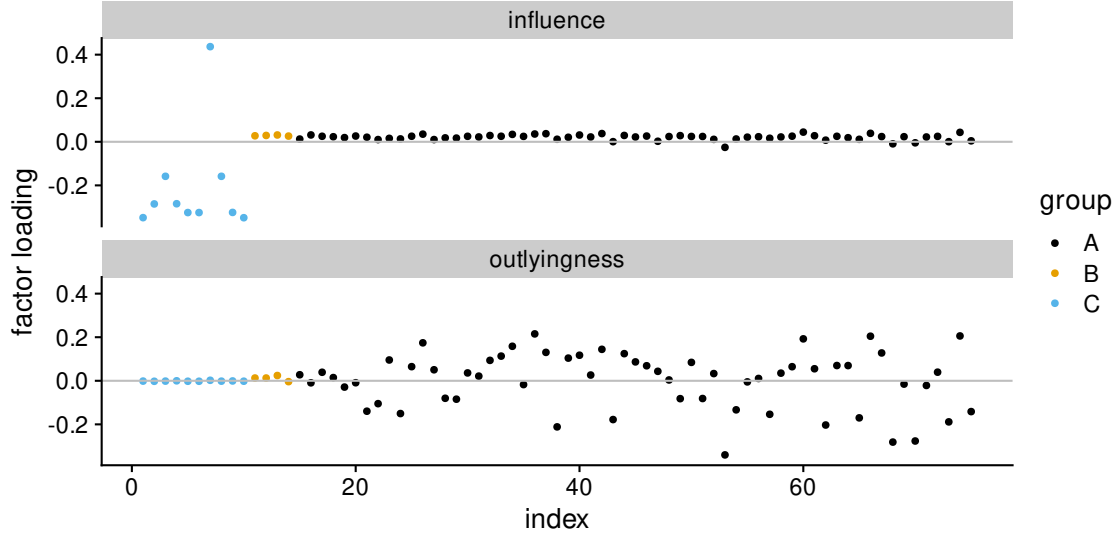


Figure 2: Diagnostics for a normal mixture model fitted to the HBK data. The top panel (influence) shows factor loadings for the first principal component of V corresponding to a maximally influential perturbation. The bottom panel (outlyingness) shows factor loadings for the first principal component of the outlier matrix Ω corresponding to a maximally outlying perturbation.

shows the maximally outlying perturbation, where observations in group C are given almost zero weight. The observations in group C are collectively influential but are not considered outliers in a mixture model that explicitly accounts for the presence of anomalous observations.

2 Cross conflict

The HBK data illustrates extreme cross-conflict. If any one of the observation groups A, B, or C is removed, then the omitted data will be in contradiction with a linear regression line fitted to the remaining two groups (See Figure 1). To quantify this, the cross-conflict ratio p_{Vi}/p_{Wi} was calculated for each group in turn. The results are shown in table 1. The top row shows the results for a linear model with a normal error distribution. This shows very high cross-conflict ratios for all groups with group A having the highest ratio with $p_{VA}/p_{WA} = 59.5$. The second row shows results for the mixture model. Cross-conflict ratios are substantially reduced compared to the linear model but still high for groups A (4.4) and C (7.9). The use of an outlier-robust mixture model does not eliminate the cross-conflict, which is a feature of the data, not the model.

Model	Group		
	A	B	C
Normal	59.5	4.89	17.6
Normal mixture	4.4	1.81	7.9

Table 1: Cross-conflict in the HBK data measured by the ratio p_{Vi}/p_{Wi} when only only outcomes from group $i \in \{A, B, C\}$ are considered as data and information the other two groups is absorbed into the prior.

References

- Hawkins, D. M., Bradu, D., and Kass, G. V. (1984). Location of several outliers in multiple-regression data using elemental sets. *Technometrics*, 26(3):197–208.
- Thomas, Z. M., MacEachern, S. N., and Peruggia, M. (2018). Reconciling curvature and importance sampling based procedures for summarizing case influence in bayesian models. *Journal of the American Statistical Association*, 113(524):1669–1683.